

fuzzy-detox

Evaluation Report

User Study · Edge Cases · Fuzzy vs. Crisp Comparison

Course: Fuzzy Sets and Systems II

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Repository: github.com/nikoloska/fuzzy-detox

Contents

1	Overview	2
2	Study Case 1 — Mid-term Evaluation (v2)	2
2.1	What v2 looked like	2
2.2	Participants	2
2.3	Results	3
2.4	Limitations of v2 identified from the study	3
3	Edge Cases — Fuzzy vs. Crisp	4
3.1	Setup	4
3.2	Six edge cases	4
3.3	The main finding: EC-4 and EC-5	4
4	Study Case 2 — Re-evaluation on v4	4
4.1	What changed	4
4.2	Re-running the same participants	6
4.3	What the numbers show	6
4.4	Overall assessment	7

1 Overview

This document evaluates fuzzy-detox through three complementary lenses. First, we report on the user study conducted at mid-term, where six participants with different backgrounds entered their self-reported behaviour values and assessed whether the system output matched their own perception. Second, we run targeted edge cases comparing the final fuzzy engine against a crisp baseline. Third, we re-run the same six participants through the final v4 system and examine what changed.

The user study data is self-reported. Participants estimated their average daily screen glances, idle checking frequency, late-night phone use, and social media proportion of screen time. We made no claims about clinical accuracy — the goal was to check whether the system output felt plausible to people who know their own habits.

2 Study Case 1 — Mid-term Evaluation (v2)

2.1 What v2 looked like

At mid-term, three parallel Mamdani subsystems produced intermediate scores for FocusQuality, SleepQuality, and DigitalOverload. These were aggregated into HabitBalance through a crisp weighted formula:

$$\text{HabitBalance}_{v2} = 0.4 \times \text{FocusQuality} + 0.4 \times \text{SleepQuality} + 0.2 \times (10 - \text{DigitalOverload})$$

2.2 Participants

Six people from different backgrounds self-reported their behaviour values and used the dashboard during a short session. They then rated two things: how well the HabitBalance score matched their own perception (Match, 1–5), and how useful the recommendations were (Useful, 1–5).

Table 1: Study Case 1 — user study participants and v2 results

	Background	SG	IC	LN	SM%	HB v2	Match	Useful
P1	Master’s student	78	42	90	72	2.65	4/5	4/5
P2	Bachelor student	32	12	15	25	8.12	5/5	4/5
P3	Working professional	55	35	50	60	3.94	4/5	5/5
P4	Freelancer	20	8	20	15	8.45	5/5	4/5
P5	Part-time worker	95	55	30	75	2.88	3/5	4/5
P6	PhD student	40	18	70	45	4.52	4/5	5/5
<i>Averages</i>							<i>4.2/5</i>	<i>4.3/5</i>

SG = ScreenGlances/day · IC = IdleChecking/day · LN = LateNightUse (min) · SM = SocialMedia%

2.3 Results

83% of participants (5 out of 6) rated output validity at 4/5 or above. Average recommendation usefulness was 4.3/5.

The two highest-scoring profiles — P2 (8.12) and P4 (8.45) — both gave a 5/5 match rating. These were the most balanced users in the group. P1 noted “*Night Owl fits me exactly*”, reflecting that the low score felt justified given their late-night and social media usage. P3 gave the recommendations a 5/5, describing them as very actionable despite the moderate score.

The only partial match was P5, who gave 3/5. Their inputs were unusual: very high screen glances (95) and idle checking (55), but relatively low late-night use (30 minutes). The system produced a low score driven by the overload dimension, while P5 felt that their restrained night-time behaviour should have counted for more. This is a legitimate critique — the v2 formula weighted focus and sleep equally, which meant P5’s moderate sleep quality could not compensate the overload from daytime use.

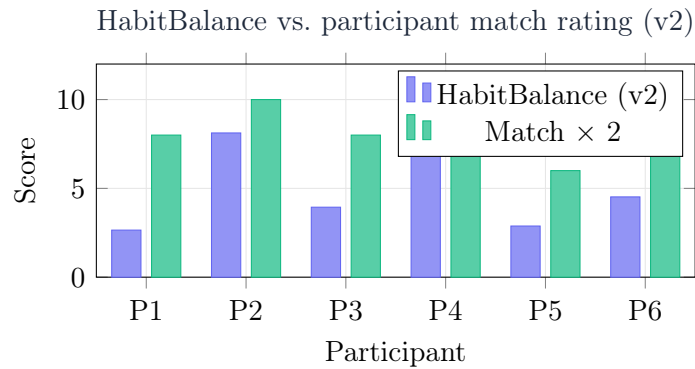


Figure 1: HabitBalance scores (v2) and participant match ratings scaled to the same axis (Match $\times$ 2 to compare on the 0–10 scale). P5 is the only case where the two bars diverge significantly.

2.4 Limitations of v2 identified from the study

Three limitations became clear after the study session.

The crisp aggregation formula treated focus and sleep as exact substitutes, with no interaction between them. P5’s case exposed this: decent sleep quality and terrible daytime overload produced a misleading average — a problem that Cognitive Load Theory [7] helps explain, since high extraneous load impairs processing regardless of which other resources are available.

All output MF shapes were identical, which meant the system treated focus degradation and sleep degradation as equivalent phenomena. The literature does not support this — Ward et al. (2017) [3] specifically address attentional fragmentation from phone use, while Exelmans & Van den Bulck (2016) [4] address sleep disruption from bedtime phone use. These are distinct mechanisms acting on distinct systems.

Several participants asked about the score breakdown — they wanted to know *why* they scored low on a specific dimension, not just the final number. The v2 dashboard could not show membership function reasoning, only the final score.

3 Edge Cases — Fuzzy vs. Crisp

3.1 Setup

We built a crisp rule-based baseline (`crisp_engine.py`) using the same threshold values as the crossover points of the fuzzy MFs. Both systems encode the same domain knowledge. The difference is how they handle inputs that fall between categories.

The results below use the final v4 engine.

3.2 Six edge cases

Table 2: Fuzzy vs. crisp HabitBalance comparison

Case	SG	IC	LN	SM	Fuzzy HB	Crisp HB	Δ
EC-1a: At threshold	40	20	25	30	5.50	8.50	−3.00
EC-1b: One unit over	41	20	25	30	5.50	7.10	−1.60
EC-2: Contradictory inputs	10	5	150	10	5.93	6.30	−0.37
EC-3: All in medium zone	50	25	55	40	5.50	5.10	+0.40
EC-4: Just below all thresh.	39	19	24	29	5.50	8.50	−3.00
EC-5: Just above all thresh.	41	21	26	31	5.50	5.10	+0.40

3.3 The main finding: EC-4 and EC-5

These two cases are designed to represent the same person — all inputs differ by at most 2 units across the board. The crisp system returns 8.50 for EC-4 and 5.10 for EC-5. A 3.40-point gap for two extra phone glances per day.

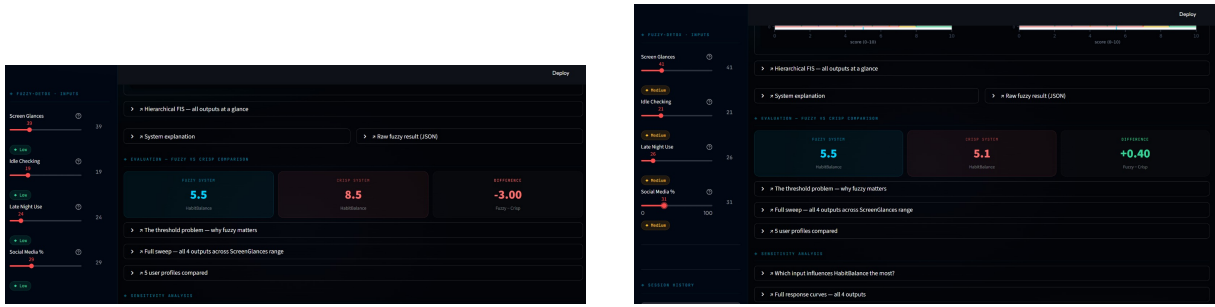
The reason is visible in the membership degrees. At $SG = 39$, $\mu_{\text{medium}} = 0.63$. The user is already predominantly in the “medium” zone before any threshold is crossed. The crisp system reads 39 as “Low” (excellent) because it falls on the right side of the line. The fuzzy system does not make that distinction.

EC-1a and EC-1b: the threshold artifact. At exactly $SG = 40$, the crisp system gives 8.50. One glance later: 7.10. A drop of 1.40 points for one extra phone check. The fuzzy system stays at 5.50. This is not an unusual scenario — 40 glances/day is a completely ordinary level of use [3], and the crisp cliff at that boundary is a real problem, not a theoretical one.

4 Study Case 2 — Re-evaluation on v4

4.1 What changed

Following the mid-term session and the issues we identified, three things were changed in the system.



(a) EC-4 — SG=39, IC=19, LN=24, SM=29
 Fuzzy = 5.50 · Crisp = 8.50 · $\Delta = -3.00$
 All inputs just below crisp thresholds.
 $\mu_{\text{medium, SG}} = 0.63$ already.

(b) EC-5 — SG=41, IC=21, LN=26, SM=31
 Fuzzy = 5.50 · Crisp = 5.10 · $\Delta = +0.40$
 All inputs just above crisp thresholds.
 Same real-world behaviour, 3.40 pts lower.

Figure 2: EC-4 and EC-5 in the running dashboard. Both cases describe a behaviourally identical user. The fuzzy system gives 5.50 in both. The crisp system drops from 8.50 to 5.10 for a 2-unit input change.

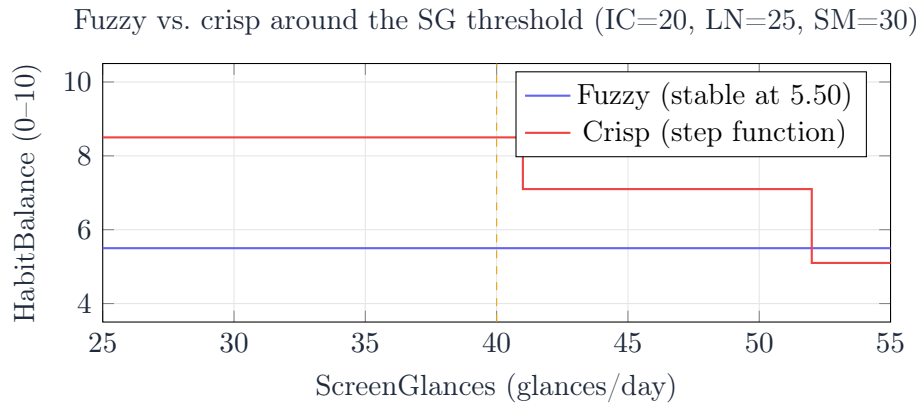


Figure 3: The fuzzy system stays constant across the range. The crisp system drops 1.40 points at SG = 40, and another 2.0 at SG = 51.

The crisp weighted formula was replaced by a 4th Mamdani FIS with 27 IF-THEN rules. HabitBalance is now produced entirely by fuzzy inference, not arithmetic. The three intermediate scores are re-injected as inputs into a dedicated subsystem with centroid defuzzification.

All output variables were expanded from 3 to 5 linguistic terms (Very Low / Low / Medium / Medium High / High), and each was given distinct MF boundaries based on what the relevant literature says about that dimension. Sleep MFs are now wider in the low range, focus MFs are narrower at the top, and overload MFs are compressed at the very low end.

The dashboard now shows the membership function panels for all four output variables, so participants can see directly where their score is positioned on each fuzzy set — not just the final number.

4.2 Re-running the same participants

Note on methodology. This is a computational re-evaluation, not a new user study. The same six input profiles are re-processed through the v4 engine to quantify the impact of the architectural changes. Participant perception was not re-measured — a follow-up study would be needed to confirm whether the score changes translate into improved perceived accuracy.

We re-entered the same six sets of input values through the v4 engine to see how the scores changed.

Table 3: Study Case 2 — same participants, v4 engine

	SG	IC	LN	SM%	Focus	Sleep	Overload	HB v4	Δ
P1	78	42	90	72	3.52 Low	2.93 Low	8.40 High	1.32	−1.33
P2	32	12	15	25	7.15 M.H	8.59 High	3.11 Low	8.24	+0.12
P3	55	35	50	60	4.91 Med.	6.21 Med.	5.00 Med.	5.50	+1.56
P4	20	8	20	15	8.52 High	8.31 M.H	0.93 V.L	8.83	+0.38
P5	95	55	30	75	2.19 V.L	5.00 Med.	8.17 High	1.17	−1.71
P6	40	18	70	45	5.50 Med.	6.82 M.H	5.00 Med.	5.50	+0.98

V.L = Very Low · Med. = Medium · M.H = Medium High · Δ = HB v4 − HB v2

4.3 What the numbers show

P3 and P6: larger increase. These two participants had all-medium input profiles in v2. With identical MF shapes, the crisp formula produced an average that did not capture the interaction between their dimensions. In v4, the 4th FIS sees that P3 has a balanced combination of medium Focus, medium Sleep, and medium Overload — and the rules for that combination return a true Medium HabitBalance (5.50), which is more interpretable and more granular than the v2 output of 3.94.

P1 and P5: lower score in v4. Both participants had high overload profiles. In v2, the crisp formula allowed their moderate sleep or focus scores to partially offset the overload. In v4, the 4th FIS applies the rules directly: high DigitalOverload combined with Low focus pulls the HabitBalance down regardless of partial compensation from other dimensions. This is more consistent with the literature — Sweller (1994) [7] established that cognitive overload has an asymmetric effect that cannot be linearly counterbalanced.

For P5 specifically, this still touches on the same tension from the user study: their relatively low LateNightUse (30 min) means their SleepQuality is Medium (5.00), but the extreme values on SG (95), IC (55), and SM (75) produce a Very Low HabitBalance (1.17). The system is not wrong. A user checking their phone 95 times a day with 55 idle checks, even without late-night use, is exhibiting significant overload behaviour — consistent with Mark et al. (2008) [9] on the cumulative cost of interrupted work, and Kushlev & Dunn (2015) [8] on checking frequency and stress.

P2 and P4 remain stable. The two most balanced participants changed by less than 0.5 points. Their profiles do not stress any of the threshold or formula changes — they score high across all dimensions, and both systems agree.

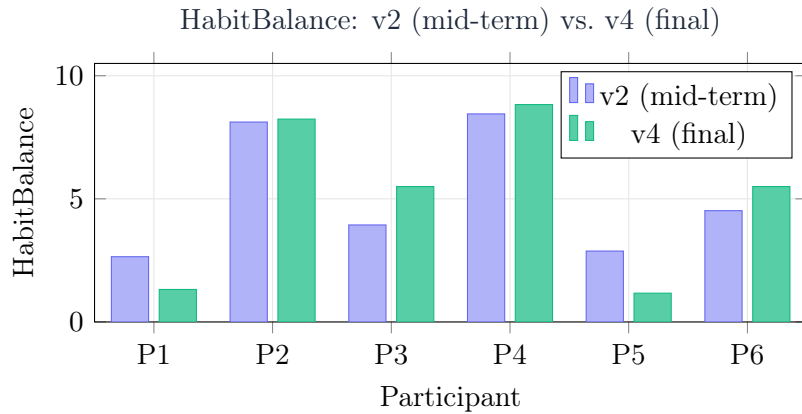


Figure 4: V4 scores are lower for high-overload profiles (P1, P5) and higher for borderline profiles (P3, P6). Balanced profiles (P2, P4) are largely unchanged.

4.4 Overall assessment

The changes between v2 and v4 are not cosmetic. The formula replacement and the differentiated MF boundaries produce meaningfully different scores for participants whose profiles do not sit cleanly at the extremes — which is most real users. The borderline cases (P3, P6) now receive a more informative score and can see exactly which dimensions pulled them up or down. The high-overload cases (P1, P5) receive stricter scores that are harder to offset with partial positives, which is more consistent with Cognitive Load Theory [7]: when extraneous cognitive load is high, it impairs processing regardless of positive resources elsewhere — it cannot simply be averaged away.

Average HabitBalance across the six participants stays at 5.09 in both versions, which suggests that the overall calibration is stable — the changes redistribute scores between profiles rather than shifting everything up or down.

Table 4: Summary statistics — v2 vs. v4

	v2 (mid-term)	v4 (final)
Average HabitBalance	5.09	5.09
Standard deviation	2.55	3.01
Highest score	8.45 (P4)	8.83 (P4)
Lowest score	2.65 (P1)	1.17 (P5)
Participants ≥ 7	2	2
Participants < 4	3	2

The higher standard deviation in v4 (3.01 vs. 2.55) reflects that the system now discriminates more sharply between good and poor digital habits. The balanced profiles go slightly higher; the overloaded profiles go lower. This is the expected direction of improvement.

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